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1. Arithmetic Operations in 8085 Microprocessors

a. 8-BIT ADDITION WITH CARRY USING DIRECT ADDRESSING AIM:

To write an assembly language program to add two numbers of 8-bit data stored in memory locations 4200H and 4201H and store the result in 4202H and 4203H with carry using direct addressing.

APPARATUS REQUIRED:

1. 8085 microprocessor kit 1

2. Power card 1

3. Keyboard 1

ALGORITHM:

1. Load the first data from memory to the accumulator and move it to B register. 2. Load the second data from memory to the accumulator.

3. Add the content of B – register to accumulator

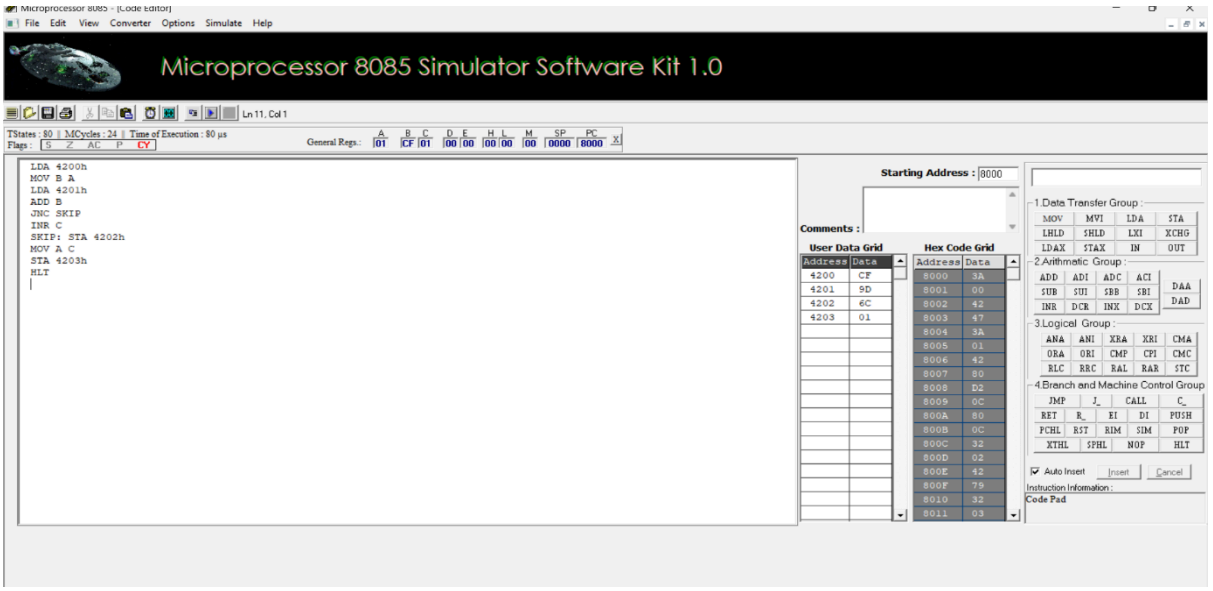
4. If Carry flag = 0 then jump to step 6

5. Increment C register to count the carry

6. Store the sum in memory.

7. Move the carry to accumulator and store in memory.

8. Stop.



RESULT:

Thus, an assembly language program for addition of two 8-bit numbers with carry was written, executed and Verified the Result successfully using 8085 kit.

b. 8-BIT ADDITION USING INDIRECT ADDRESSING

AIM:

To write an assembly language program to add two numbers of 8-bit data stored in memory locations 4200H and 4201H and store the result in 4202H and 4203H using indirect addressing modes

APPARATUS REQUIRED:

1. 8085 microprocessor kit 1

2. Power card 1

3. Keyboard 1

ALGORITHM:

1. Load the first data from memory to the accumulator.

2. Add the content of Memory to accumulator

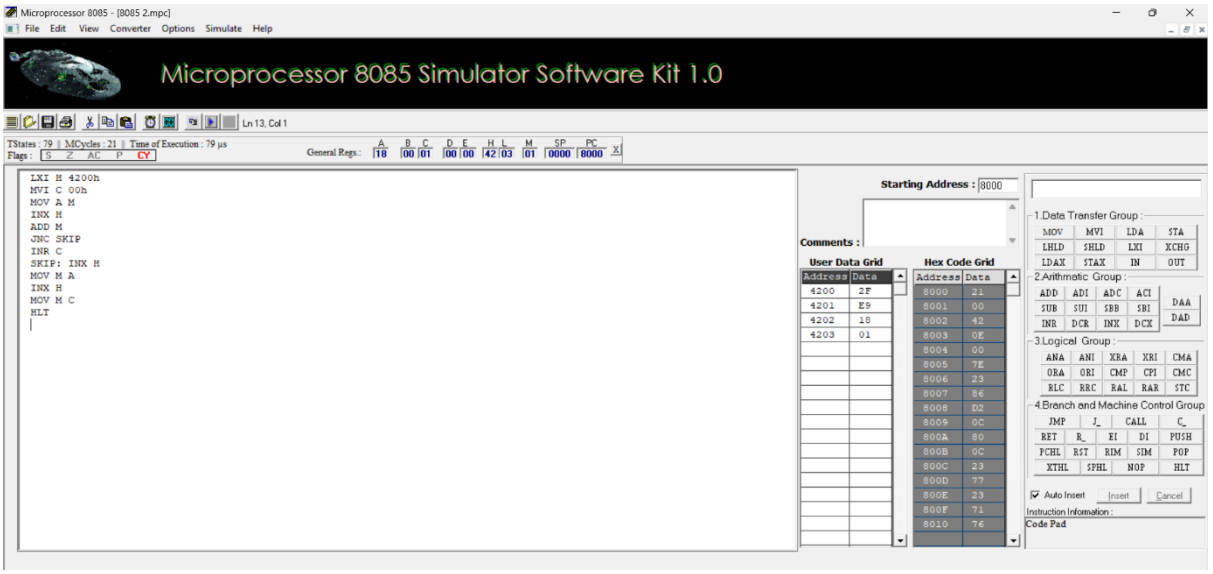
3. If Carry flag = 0 then jump to step 5

4. Increment C register to count the carry

5. Store the sum in memory.

6. Store the carry in memory from C-register.

7. Stop.



RESULT:

Thus, an assembly language program for addition of two 8-bit numbers with carry was written, executed and Verified the Result successfully using 8085 kit.

c. 8-BIT SUBTRACTION WITHOUT BORROW

AIM:

To write an assembly language program to subtract two numbers of 8-bit data stored in memory 4200H and 4201H. Store the magnitude of the result in 4202H using 8085 Microprocessor.

APPARATUS REQUIRED:

1. 8085 microprocessor kit 1

2. Power card 1

3. Keyboard 1

ALGORITHM:

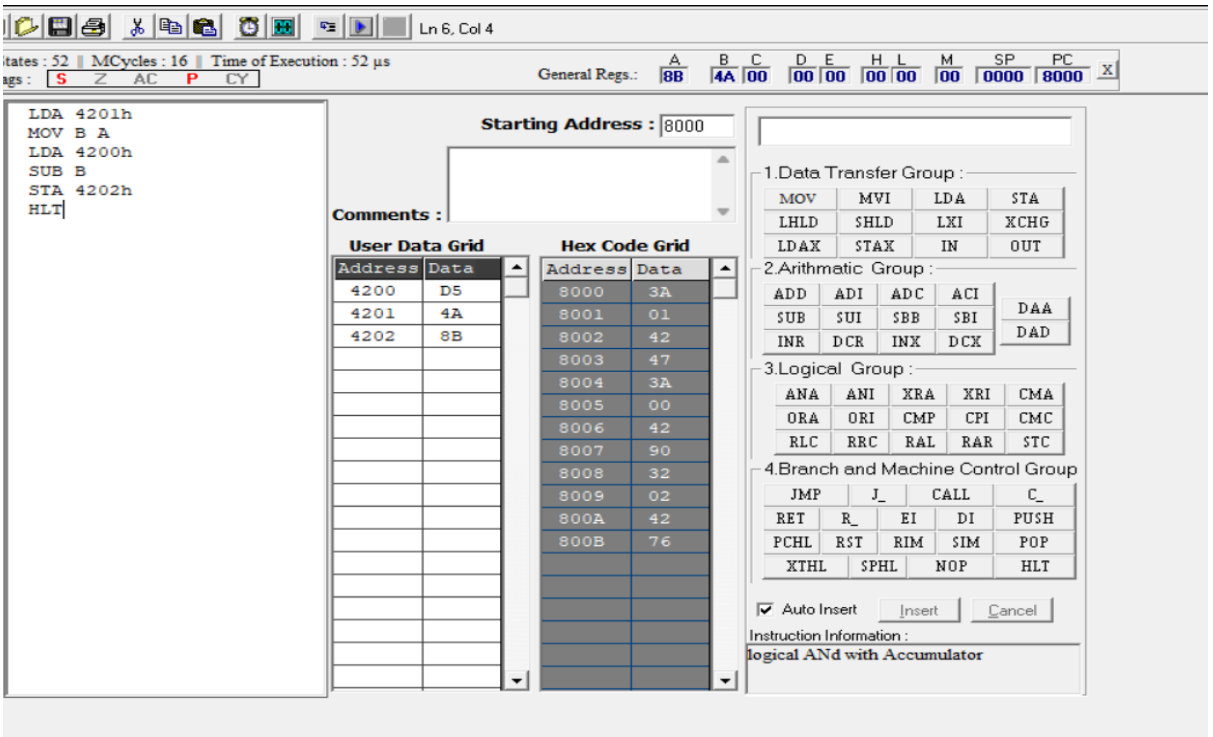
1. Load the subtrahend (the data to be subtracted) from memory to accumulator and move if to B-register.

2. Load the minuend from memory to accumulator.

3. Subtract the content of B-register (subtrahend) from the content of accumulator (minuend).

4. Store the difference in memory.

5. Stop



RESULT:

Thus, an assembly language program for subtraction of two 8-bit numbers without borrow was written, executed and Verified the Result successfully using 8085 kit.

d. 8-BIT SUBTRACTION WITH BORROW USING DIRECT ADDRESSING

AIM:

To write an assembly language program to subtract tow numbers of 8-bit data stored in memory locations 4200H and 4201H and store the result in 4202H and 4203H with borrow using direct addressing.

APPARATUS REQUIRED:

1. 8085 microprocessor kit 1

2. Power card 1

3. Keyboard 1

ALGORITHM:

1. Load the second data from memory to the accumulator and move it to the B register. 2. Load the first data from memory to the accumulator.

3. Subtract the content of B – register from accumulator

4. If Carry flag = 0 then jump to step 5 & 6

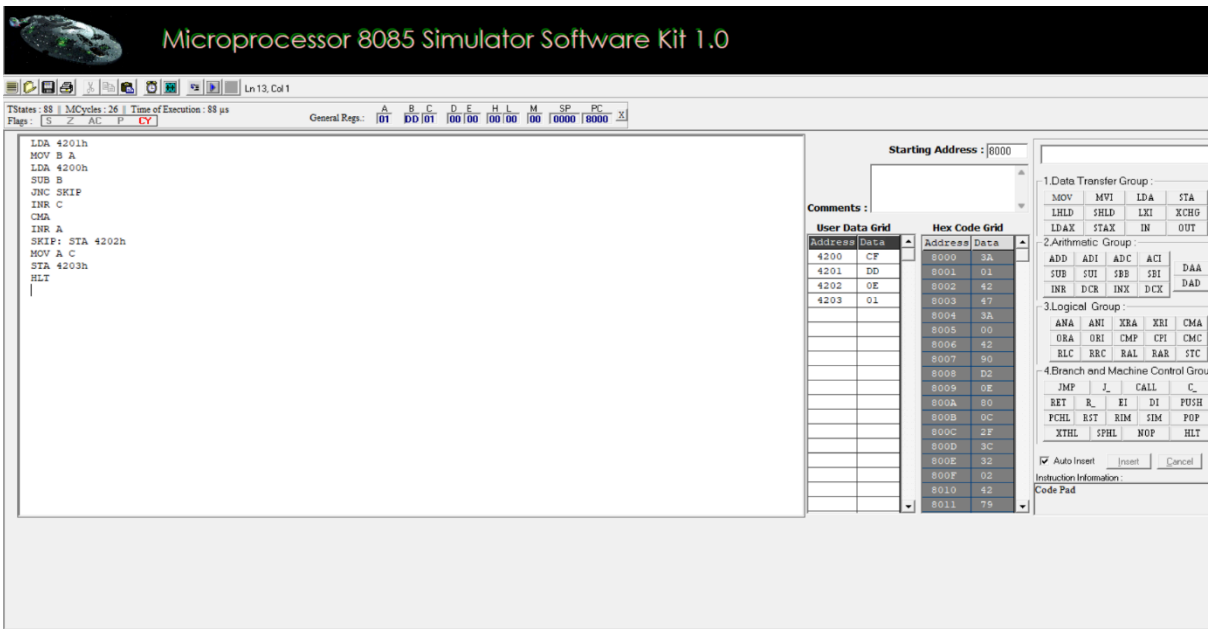
5. Increment C register to count the borrow

6. Take two's complement of the difference

7. Store the Difference in memory.

8. Move the borrow to accumulator and store in memory.

9. Stop.



RESULT:

Thus, an assembly language program for subtraction of two 8-bit numbers with borrow was written, executed and Verified the Result successfully using 8085 kit.

e. 8-BIT MULTIPLICATION OPERATIONS USING 8085 MICROPROCESSOR

AIM:

To write an assembly language program to multiply two numbers of 8-bit data stored in memory 4200H and 4201H and store the product in 4202H and 4203H.

APPARATUS REQUIRED:

1. 8085 microprocessor kit 1

2. Power card 1

3. Keyboard 1

ALGORITHM:

● Load the first data in ACC and move to E.

● Load the second data ACC and move to B (count)

● Clear HL pair (Initial sum)

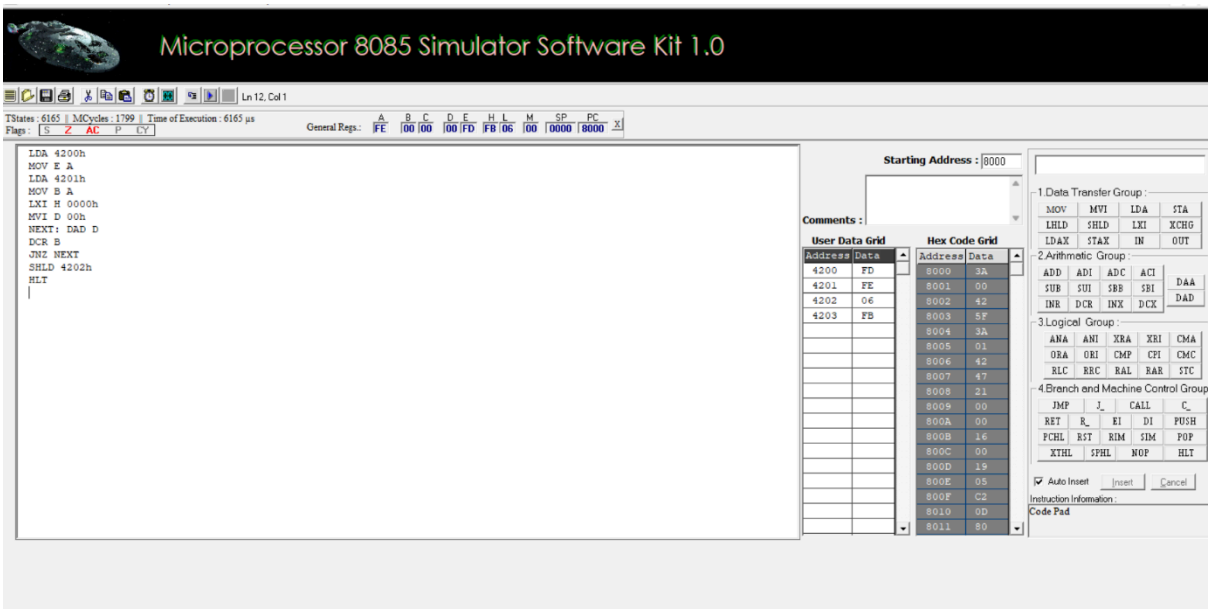
● Clear D for overflow (carry)

● Add the content of DE to HL

● Decrement the count.

● Check whether the count has reached zero.

● Check the zero flag. If ZF = 0, repeat addition or If ZF = 1, go to next step ● Store the content of HL in memory. (Least significant 16 bits of the product) ● Stop.



RESULT:

Thus, an assembly language program to multiply two numbers of 8-bit data was written, executed and Verified the Result successfully using 8085 kit.

f. 8-BIT DIVISION OPERATIONS USING 8085

MICROPROCESSOR

AIM:

To write an ALP to perform division of two 8 bit numbers Stored in memory location 4200H, 4201 H and Store the remainder in 4202H and the quotient in 4203H.

APPARATUS REQUIRED:

1. 8085 microprocessor kit 1

2. Power card 1

3. Keyboard 1

ALGORITHM:

● Load the divisor in accumulator and move if to B-register

● Load the dividend in the accumulator.

● Clear C-register to account for quotient

● Check whether divisor is less than dividend

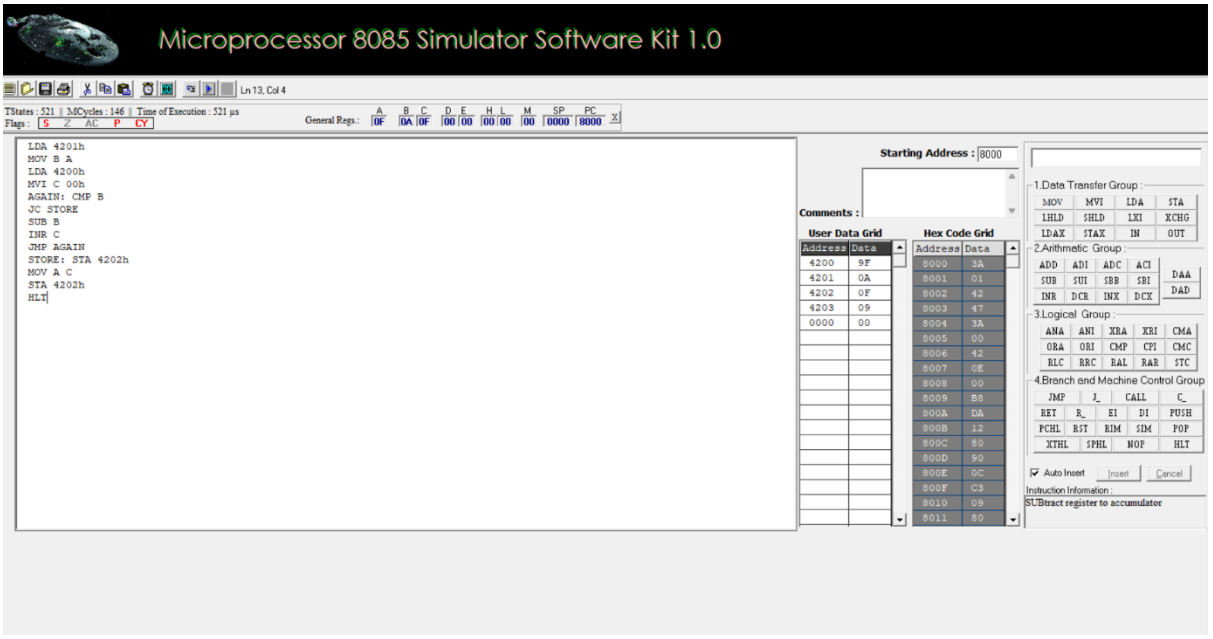
● If divisor is less than dividend, go to step 8, otherwise go to next step ● Subtract the content of B-register (quotient)

● Increment the content of C-register (quotient)

● Go to step 4

● Store the content of the accumulator (remainder) in memory.

● Move the content of C-register (quotient) to accumulator and store in memory ● Stop.



RESULT:

Thus, an assembly language program to Divide two numbers of 8-bit data was written, executed and Verified the Result successfully using 8085 kit.